

Unit 4, Lesson 1: Rational Numbers and the Number Line – Part 1



Benchmark

MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context.



Learning Targets

- ☐ I can demonstrate an understanding of negative numbers.
- ☐ I can recognize that a number in the real number system can have a variety of classifications.
- ☐ I can plot rational numbers on a horizontal and vertical number line.



4.1.1 Warm-Up: Creating Zero

1. Use only digits 1 to 9, at most one time each, so that the equation is true.

$$(\square + \square - \square + \square) - (\square + \square - \square + \square) = 0$$

2. Switch your answer with a partner to check. Ask your partner his/her strategy for finding his/her answer and write his/her strategy below.



4.1.2 Exploration: How Can Negative Numbers Exist?

Negative numbers are numbers less than zero.

Examples include:

- a temperature below 0 degrees
- the amount of money owed, or debt
- a contestant's score on Jeopardy (when someone misses a question, he/she loses points, so his/her score could be a negative amount)

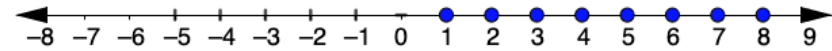
1. Discuss the examples of negative numbers with your partner.
2. Discuss with your partner which of the contexts in the table could have a value less than 0, then write an example or explain your thinking.

Context	Example or Explanation
Direction	
Points in a game	
Time	



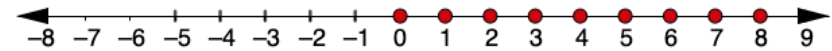
4.1.3 Guided Instruction: The Real Number System

The values in blue on the number line are some of the numbers in the set of **natural numbers**.



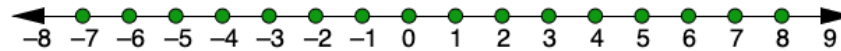
Natural numbers are the counting numbers
 $\{1, 2, 3, 4, 5 \dots\}$.

The values in red on the number line are some of the numbers in the set of **whole numbers**.



Whole numbers are natural numbers and zero.
 $\{0, 1, 2, 3, 4, 5 \dots\}$

The values in green on the number line are some of the values in the set of **integers**.



Integers are whole numbers and their opposites.
 $\{\dots -4, -3, -2, -1, 0, 1, 2, 3, 4 \dots\}$

1. Use the number lines to help to answer each of the following questions.
 - a. What do you notice about the set of natural numbers and the set of whole numbers?
 - b. What do you notice about the set of integers?
 - c. What type of number is -10 ?
 - d. What type of number is 27 ?
 - e. Describe the numbers that appear in all three sets.

Camaria wondered to herself, "What about numbers such as $3\frac{1}{2}$ and -1.75 ?"

These numbers are examples of **rational numbers**.



Rational numbers are numbers that can be expressed as the ratio of two integers.

2. Use the definition of rational numbers to write each as a ratio of two integers.

Number	Ratio of Two Integers
-1.5	
9	

In your study of mathematics, you will learn about **irrational numbers**, which are numbers that cannot be written as a ratio of two integers.



The set of **real numbers** is the set of all rational and irrational numbers.



4.1.4 Activity: Horizontal Number Line

Using a sheet of paper, follow the steps to make your own number line.

Step 1: Use a straightedge or a ruler to draw a horizontal line. Mark the middle point of the line and label it 0.

Step 2: To the right of 0, draw tick marks that are 1 centimeter apart. Label the tick marks 1, 2, 3...10.

Step 3: Fold your paper so that a vertical crease goes through 0 and the two sides of the number line match up perfectly.

Step 4: Use the fold to help you trace the tick marks that you already drew onto the opposite side of the number line. Unfold and label the tick marks $-1, -2, -3 \dots -10$. This represents the negative side of your number line.

Use the number line to answer each of the following questions.

1. Describe how far away the number 5 is from 0.

The number 5 is _____ units to the _____ of 0 on the number line.

2. Write a positive number and a negative number where each is farther away from 0 than the number 5.

3. Describe how far away the number -2 is from 0.

The number -2 is _____ units to the _____ of 0 on the number line.

4. Write a positive number and a negative number that are each farther away from 0 than -2 .



4.1.5 Bring It Together: Plotting Rational Numbers on Number Lines

When plotting numbers on a number line, the sign of the number indicates which direction from 0 to move on the number line.

- On a horizontal number line, positive numbers are to the

☐ right

☐ left

 of zero.
- On a vertical number line, positive numbers are

☐ above

☐ below

 zero.
- On a horizontal number line, negative numbers are to the

☐ right

☐ left

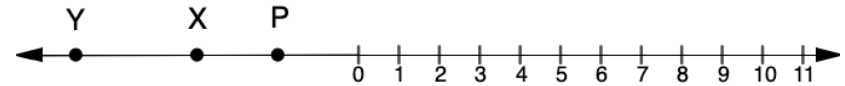
 of zero.
- On a vertical number line, negative numbers are

☐ above

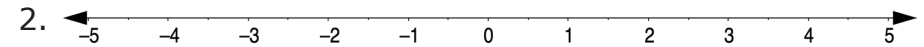
☐ below

 zero.

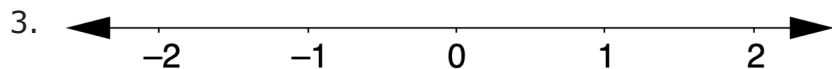
1. A number line is shown with points P , X , and Y .



- What type of numbers are P , X , and Y ?
- Estimate the location of points P , X , and Y .



- Plot the values $-3\frac{2}{3}$ and 4.3 on the number line.
- How did the tick marks between each integer help you determine where to plot 4.3?
- How did the tick marks between each integer help you determine where to plot $-3\frac{2}{3}$?

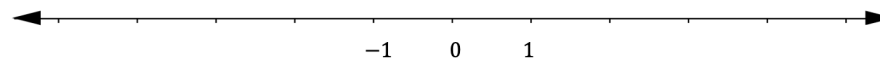


- On the number line, make a tick mark that is halfway between each value on the number line. Label each tick mark with the value of its location.
- Make another tick mark halfway between each value and the tick marks that were previously made. Label each with the value of its location.
- Plot the values $-1\frac{7}{12}$ and 0.25 on the number line.

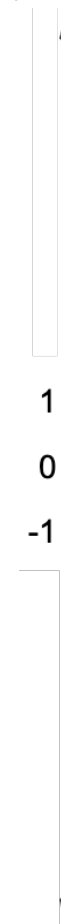


4.1.6 Try It: Plotting Rational Numbers on a Number Line

- Plot the numbers $-\frac{5}{2}$, -4 , $\frac{1}{4}$, and 3 on the horizontal number line.



- Plot the rational numbers -0.25 , -1.8 , 6.2 , and 4.75 on the vertical number line.

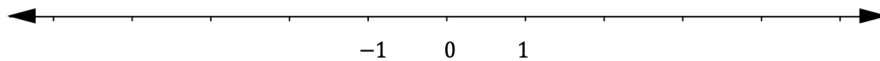




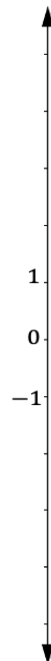
4.1.7 Your Turn!

1. Negative numbers are ☐ greater ☐ less than zero. On a horizontal number line, negative numbers are to the ☐ right ☐ left of zero. On a vertical number line, negative numbers are ☐ above ☐ below zero.

2. Plot the integers 2, -3, 0, -5 and 5 on the horizontal number line.

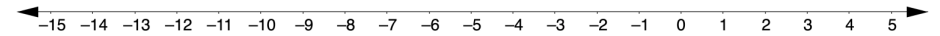


3. Plot the rational numbers $-1\frac{1}{5}$, $-\frac{4}{5}$, 2, and $\frac{6}{5}$ on the vertical number line.



4.1.8 Wrap-Up: Ticket Out the Door

1. Plot the rational numbers $\frac{-25}{4}$, $-12\frac{3}{4}$, $\frac{12}{4}$, and $\frac{1}{4}$ on the number line.



2. Explain how to estimate the value of G .

